

Reducing Cardiovascular Disease in Indonesia

Health Impact, Costs, and Cost-Effectiveness of Clinical and Public-Health Interventions, 2025–2050

World Bank · University of Washington

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The challenge, and the opportunity

Most premature cardiovascular deaths in Indonesia can be prevented or treated

Cardiovascular disease is Indonesia's largest health burden

Under current trends, a person reaching age 30 faces roughly a **14.6% chance** of dying from cardiovascular disease before age 70. Most of this is **preventable or treatable**.

- Premature CVD deaths (ages 30–69) have risen for two decades and remain high.
- Two complementary levers:
 - **Clinical care** — prevent and better treat established disease.
 - **Population prevention** — tobacco, salt, trans-fat.
- This analysis weighs their **health impact and value for money** together.

Source: Indonesia CVD model, calibrated to GBD 2023 and UN World Population Prospects 2024; combined results workbook (annual probability-of-death sheet). Method: the probability of death from CVD before age 70 is the chance of dying from the six modelled cardiovascular diseases between exact ages 30 and 70, read from a single-year life table in the 2025 baseline year.

We measured the health gains and the value for money of proven interventions

General aim

Estimate the **health impact and value for money** of scaling up evidence-based **clinical** and **public-health** cardiovascular interventions in Indonesia through **2050**.

Aim 1 — Burden

Project the CVD burden and the probability of death from CVD before age 70 under the **current course** and under **alternative intervention scenarios**.

Aim 2 — Value for money

Estimate **deaths averted**, mortality reductions, **additional costs**, and the **cost per death averted** for each intervention and package.

Aim 3 — Better together

Quantify the **added value of combining** clinical and public-health action, modelled as a single joint run rather than a sum.

***Source:** study design; Indonesia CVD model. **Method:** every scenario runs through one projection over 2025–2050; the baseline (current course) is the no-new-intervention comparator against which each scenario is measured.*

Together, clinical care and prevention could avert about 5.1 million deaths by 2050

Combining clinical + public health is the big prize

Together they could avert about **5.1M deaths** over 2025–2050, cut the probability of death from CVD before age 70 in 2050 by **31%**, at about **US\$19,299 per death averted**.

- **All clinical interventions:** $\approx$ 2.9M deaths averted; probability of death from CVD before age 70 down 21% by 2050.
- **All public-health policies:** $\approx$ 2.4M deaths averted at about US\$207 per death averted.
- **Most impactful single levers:** **Primary CVD prevention** and **Tobacco tax**.
- **Affordability:** the combined package costs about US\$19,299 per death averted.

Source: results contract (09_deck_results.rds) + combined workbook. Method: deaths averted cumulative 2025–2050 (six CVD causes + type-2 diabetes); cost per death averted = discounted additional cost $\div$ deaths averted (2023 US\$). Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

The model follows how prevention and treatment change disease and survival

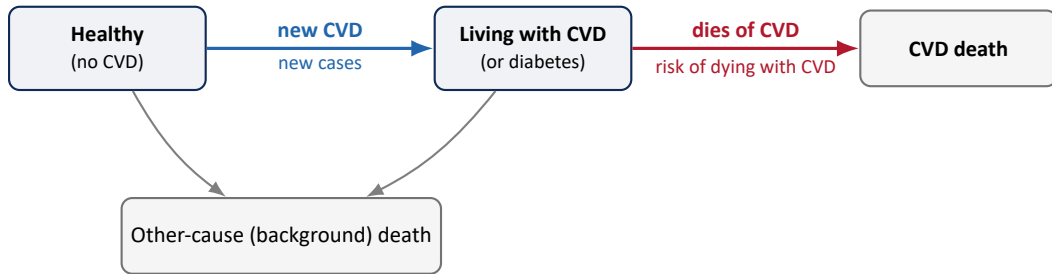
1. **People & demography.** Indonesia's population is projected by age and sex (UN World Population Prospects 2024).
2. **Starting health.** Disease and death rates are set to Indonesia's data (GBD 2023) and calibrated to match observed levels.
3. **Year-by-year simulation.** Each year people can develop cardiovascular disease or diabetes, survive, or die.
4. **Interventions.** Clinical and public-health scenarios change how many people develop disease (new cases) and how many survive it.
5. **Outcomes.** The model reports deaths, healthy years lost or gained (DALYs), the probability of death from CVD before age 70, costs, and value for money.

One consistent engine

Every scenario runs through the **same** projection, so results are directly comparable and the **combined** package is modelled as one run — not a sum of separate estimates.

Source: Indonesia CVD model engine; GBD 2023 (starting rates); UN World Population Prospects 2024 (demography, life expectancy). Method: a discrete-time state-transition simulation by single-year age and sex, calibrated to match observed GBD levels using a standard numerical optimiser.

Interventions work by preventing new disease and by helping people survive it



Fewer people fall ill

Primary prevention & *all* public-health policies reduce **new cases**.

More people survive

Secondary prevention, treatment & tobacco policy lower the **risk of dying after developing CVD**.

Background mortality

Other-cause deaths from demography; unchanged by these interventions.

Source: Indonesia CVD model engine (06_run_scenarios_indonesia_fair.R). Method: interventions act on two transitions — new cases (Healthy → Living with CVD) and the risk of dying after developing CVD (Living with CVD → death); no recovery/remission is modelled, a conservative assumption.

The interventions we tested

Clinical care prevents disease and helps people survive it

Intervention	Target population	Model effect
Primary CVD prevention	Adults at cardiovascular risk	Fewer new cases
Type-2 diabetes treatment	People with type-2 diabetes	Better survival
Heart-attack secondary prevention	Heart-attack (IHD) survivors	Better survival
Chronic heart-failure care	Chronic heart-failure patients	Better survival
Acute heart-failure care	Acute heart-failure patients	Better survival
Stroke secondary prevention	Stroke survivors	Better survival
RHD secondary prevention	Rheumatic heart disease patients	Better survival
RHD cardiac surgery	Target patient group	Better survival

Implementation scenario (all clinical interventions): coverage scales up **from a low base toward 80%** of the people who need each service between 2025 and 2050. Primary prevention lowers who develops CVD (new cases); secondary prevention and treatment improve survival.

Source: clinical results workbook, sheets SeLected_Interventions and Scenario_CatALog. Method: the table lists the clinical interventions active in this run (excluded interventions never appear); effect sizes come from the FAIR Choices intervention taxonomy.

Population policies cut tobacco, salt, and trans-fat risk across the whole population

Policy	Risk factor	Model effect	Implementation lever
Tobacco tax	Tobacco	New cases + survival	Raise tax to 75% of retail price
Sodium policy package	Dietary salt/sodium	New cases	Reformulation, labelling, media, procurement
Tobacco policy package	Tobacco	New cases + survival	Clean-air law, mass media, ad ban

Policies reduce exposure (smoking, salt, trans-fat intake), lowering the number of **new CVD cases**. **Tobacco** policies additionally improve **survival** after CVD. Packages are modelled as single joint runs of their component measures.

Source: public-health results workbook, sheets `Selected_Interventions / Policy_Levers` and `Scenario_Catalog`.

Method: the table lists the policies active in this run; effect sizes come from published risk-factor evidence stored in the workbook.

Every input is traceable to a recognised data source

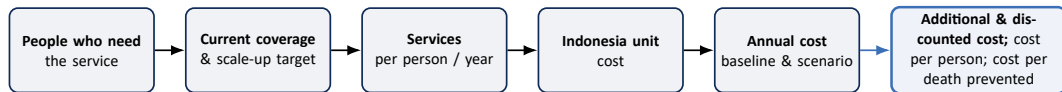
Model component	Primary source	Years / version	Use in model
Population & demography	UN World Population Prospects	2024 revision	Population, background mortality, life expectancy
Cause-specific mortality & disease rates	GBD (Indonesia)	2023	Baseline burden & calibration targets
Disability weights (healthy years lost)	GBD	2023	Healthy years lost or gained (DALYs)
Clinical intervention effects	FAIR Choices intervention taxonomy	current	New-case & survival effects
Public-health effect sizes	Published risk-factor evidence (in workbook)	current	New-case & survival effects
Intervention & policy costs	Programme & policy unit costs (in workbook)	price year 2023	Budget impact & value for money
International CVD mortality comparison	WPP-scaled ASEAN comparison series	1995–2050	Benchmarking Indonesia vs. neighbours

Population: UN World Population Prospects 2024; epidemiology: GBD 2023 (GBD 2023 Cardiovascular Diseases and Risks Collaborators 2025; Schumacher et al. 2024).

Source: modelling pipeline and input workbooks. Method: these inputs drive the demographic projection, calibration targets, and effect sizes; effect-size and cost citations are in the workbooks.

Costs depend on who needs care, how many receive it, and local prices

We estimate the **additional health-system resources required to expand effective coverage**, using a costing approach adapted from Disease Control Priorities (DCP) (Watkins et al. 2020).



What this tells policymakers: the analysis estimates how much additional health-system spending is required to move from today's coverage toward universal, high-quality access—and relates that investment to the health gains achieved.

Source: intervention-based costing adapted from Watkins et al. (2020); unit costs and coverage from the input workbooks; demography from UN WPP 2024. **Method:** $\text{annual cost} = \text{people in need} \times \text{coverage} \times \text{annual service frequency} \times \text{Indonesia-adjusted unit cost}$; $\text{additional cost} = \text{scenario} - \text{baseline}$; equations in the appendix. **Preliminary costing results:** unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

What the interventions deliver, and at what cost

Scaling up clinical care sharply cuts premature cardiovascular deaths

Clinical intervention	Deaths averted (cumulative)	Prob. of CVD death before age 70, 2050	Reduction in that probability, 2050	Deaths reduction 2050	Cost per death averted	Add'l cost/capita per yr, undisc.	Add'l cost/capita per yr, disc.
Primary CVD prevention	1,964,000	9.7%	16.7%	11.5%	US\$4,998	US\$2.02	US\$1.25
Type-2 diabetes treatment	493,000	11.6%	0.6%	2.5%	US\$160,079	US\$16.42	US\$10.00
Heart-attack secondary prevention	244,000	11.4%	2.4%	1.1%	US\$20,605	US\$1.05	US\$0.64
Chronic heart-failure care	105,000	11.6%	0.9%	0.4%	US\$9,010	US\$0.20	US\$0.12
Acute heart-failure care	94,000	11.6%	0.6%	0.4%	US\$14,665	US\$0.29	US\$0.18
Stroke secondary prevention	40,000	11.7%	0.3%	0.2%	US\$70,561	US\$0.58	US\$0.36
RHD secondary prevention	2,000	11.7%	-0.3%	0.0%	US\$33,026	US\$0.01	US\$0.01
RHD cardiac surgery	0	11.7%	0.0%	0.0%	US\$2,748,217	US\$0.17	US\$0.11
All clinical (combined)	2,884,000	9.3%	20.6%	15.7%	US\$34,063	US\$20.40	US\$12.46

Source: results contract (09_deck_results.rds); Model 09 divides each year's additional cost by that year's Indonesia population from model_output_Indonesia_htncov2_aspirational.rds (workbooks carry the same as live Budget_Impact / Cost_Effectiveness formulae). Method: cost per death averted = discounted additional cost ÷ deaths averted (3%/yr, 2023 US\$, health system); the two per-capita columns are the simple mean of the annual additional cost per capita (undiscounted / discounted) over 2026–2050. Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

Population policies avert deaths at very low cost

Public-health policy	Deaths averted (cumulative)	Prob. of CVD death before age 70, 2050	Reduction in that probability, 2050	Deaths reduction 2050	Cost per death averted	Add'l cost/capita per yr, undisc.	Add'l cost/capita per yr, disc.
Tobacco tax	1,082,000	11.0%	5.9%	3.7%	US\$189	US\$0.04	US\$0.03
Sodium policy package	772,000	11.2%	4.0%	2.9%	US\$233	US\$0.03	US\$0.02
Tobacco policy package	650,000	11.3%	3.6%	2.2%	US\$182	US\$0.02	US\$0.02
All public health (combined)	2,429,000	10.2%	13.0%	8.6%	US\$207	US\$0.10	US\$0.07

Source: results contract (09_deck_results.rds); Model 09 divides each year's additional cost by that year's Indonesia population from model_output_Indonesia_htncov2_aspirational.rds (workbooks carry the same as live Budget_Impact / Cost_Effectiveness formulae). Method: cumulative deaths averted 2025–2050 vs baseline; tobacco and sodium packages each modelled as one joint run; the two per-capita columns are the simple mean of the annual additional cost per capita (undiscounted / discounted) over 2026–2050. Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

Clinical and population action, combined

Clinical care and prevention achieve more together than alone

Package	Deaths averted (cumulative)	Reduction in prob. of CVD death <70, 2050	Prob. of CVD death before age 70, 2050	Discounted total cost	Cost per death averted	Add'l cost/capita per yr, undisc.	Add'l cost/capita per yr, disc.
All clinical (combined)	2,884,000	20.6%	9.3%	US\$98.2B	US\$34,063	US\$20.40	US\$12.46
All public health (combined)	2,429,000	13.0%	10.2%	US\$503.8M	US\$207	US\$0.10	US\$0.07
Clinical + public health (combined)	5,052,000	30.7%	8.1%	US\$97.5B	US\$19,299	US\$20.23	US\$12.37

The combined package is modelled as one run — not a sum

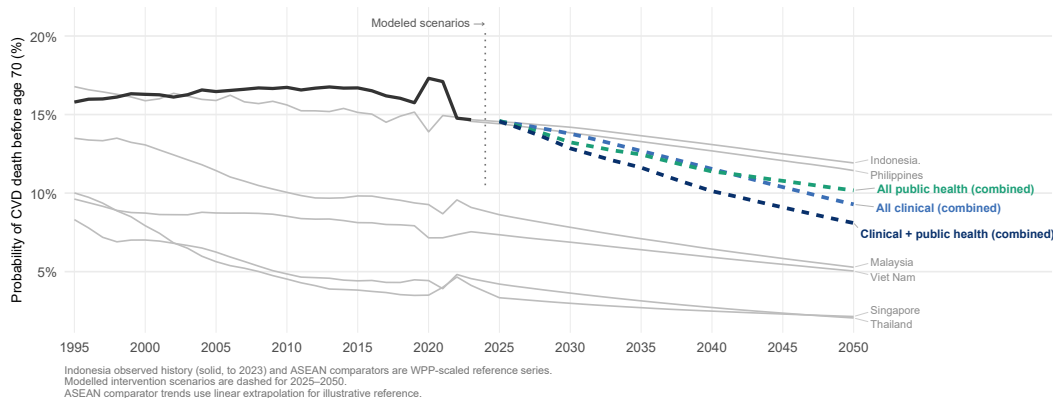
Because the two families' effects overlap, the joint scenario is projected in a single run: about **5.1M deaths** averted and the 2050 probability of CVD death before age 70 cut to **8.1%** (from 11.7%), at about **US\$19,299 per death averted**.

Source: results contract (09_deck_results.rds); Model 09 divides each year's additional cost by that year's Indonesia population from model_output_Indonesia_htncov2_aspirational.rds (workbooks carry the same as live Budget_Impact / Cost_Effectiveness formulae). Method: joint package = one projection applying both families (not their sum); the two per-capita columns are the simple mean of the annual additional cost per capita (undiscounted / discounted, 3%/yr, 2023 US\$) over 2026–2050. Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

Indonesia's premature cardiovascular mortality can fall toward its neighbours

Probability of death from CVD before age 70: observed, projected, and neighbours

Chance of dying from CVD between exact ages 30 and 70



Source: combined results workbook (annual probability-of-death sheet, R-reconciled values); ASEAN comparator series in data/raw/. **Method:** model scenarios (dashed) run 2025–2050 for the six modelled cardiovascular causes over exact ages 30–70; comparators are WPP-scaled reference series shown for context only.

A stronger care cascade: the 70-30-30 to 70-70-70 strategy

The cascade scale-up is ambitious but staged

Hypertension and type-2 diabetes are modelled with **diagnosed** → **treated** → **controlled** care cascades; **cholesterol** treatment coverage follows hypertension coverage. Coverage rises in two stages, then holds.

Two staged milestones

- **2030 — 70-30-30:** 70% diagnosed, 30% of those treated, 30% of those controlled.
- **2040 — 70-70-70:** 70% diagnosed, 70% treated, 70% controlled; held through 2050.
- Linear scale-up from the observed baseline; coverage already above a milestone is **not scaled down**.

Shares of all people with the condition

	2030	2040
Diagnosed	70%	70%
Treated	21%	49%
Controlled	6.3%	34.3%

treated = 70% × {30, 70}%; controlled = treated × {30, 70}%.

Effective modelled coverage = controlled + $\frac{1}{2} \times$ treated-but-uncontrolled = **13.65%** (2030) and **41.65%** (2040): full effect to controlled patients, half to treated-but-uncontrolled.

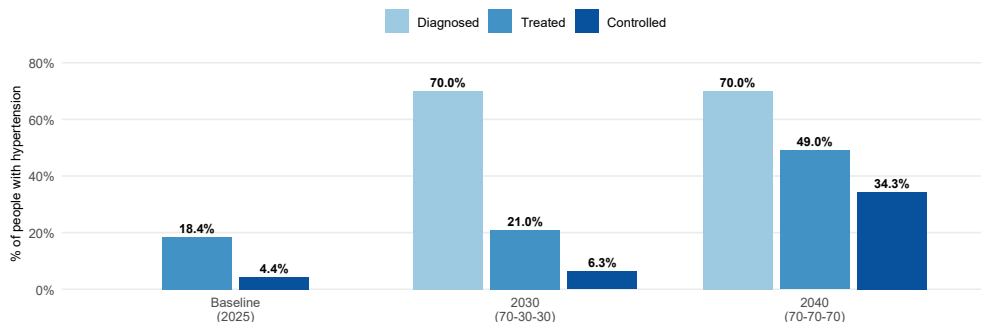
Source: cascade workbook (Cascade_Assumptions, Cascade_Trajectory); isolated runner; NCD-RisC coverage inputs.

Method: nested targets on the diagnosed/treated/controlled shares; piecewise-linear scale-up, no backsliding.

Indonesia's hypertension cascade can progress in two achievable stages

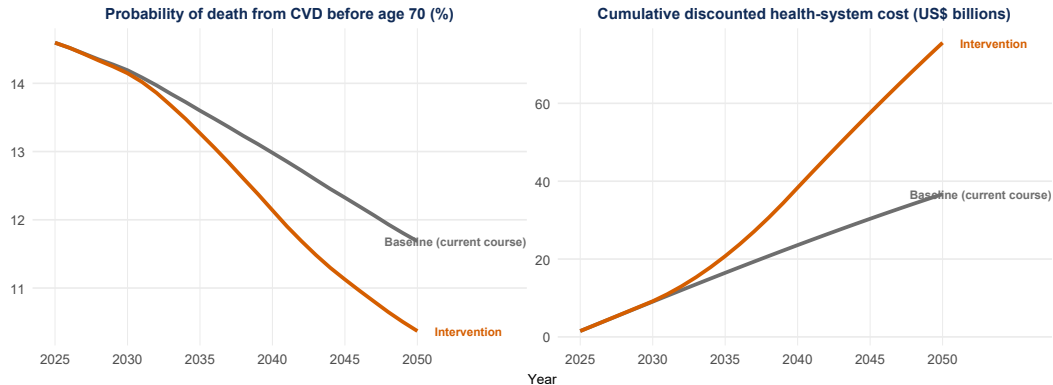
Hypertension care cascade under the 70-30-30 to 70-70-70 scale-up

Diagnosed, treated, and controlled shares of all people with hypertension (intervention scenario)



Source: *cascade Cascade_Trajectory (sex-averaged) and Cascade_Assumptions; scale-up begins 2026. Method:* baseline (2025) shows only measured treated/controlled — diagnosis is not modelled at baseline, so no diagnosed bar. Type-2 diabetes follows the same targets (baseline treated $\approx$ 22.0%, controlled $\approx$ 7.5%).

Stronger prevention lowers premature deaths at a rising but bounded cost



Source: cascade workbook — annual probability-of-death sheet (R-reconciled) and AnnualL_Cost, cumulatively summed and reconciled to Budget_Impact. **Method:** probability over the six CVD causes, ages 30–70; cumulative discounted health-system cost (3%/yr to 2025, 2023 US\$) of the CVD primary-care and type-2 diabetes cascades. **Preliminary costing results:** unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

A stronger cascade could prevent about 1.6 million deaths by 2050

Cumulative from 2025 through year	2030	2040	2050
Projected deaths — CVD (six causes) + type-2 diabetes (million)			
Baseline (current course)	6.106	18.396	33.345
Intervention	6.100	18.002	31.768
Projected health-system cost (US\$ billion, discounted)			
Baseline (current course)	9.06	23.58	36.67
Intervention	9.16	38.34	75.57
Net effect of the strategy			
Additional cost (US\$ billion, discounted)	0.11	14.76	38.90
Deaths prevented (million)	0.006	0.394	1.577

Source: cascade Annual_Mortality and Annual_Cost/Budget_Impact; scale-up begins 2026. Method: each column cumulative from 2025 through that year; deaths prevented = baseline — intervention (six CVD causes + type-2 diabetes); additional cost = intervention — baseline (discounted 3% to 2025, 2023 US\$). Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPIS and Ministry of Health data before policy use.

What this means for policy

Prevention and clinical care together offer the biggest, most affordable gains

- **Biggest health gains come from doing both.** The combined package averts far more (~5.1M) than either family alone.
- **Public-health policies are exceptional value.** Population-wide fiscal/regulatory measures avert deaths at a fraction of clinical cost per death averted.
- **Clinical scale-up delivers large, durable gains** in survival and premature mortality.
- **Sequence for early wins.** Tobacco taxation and other prevention policies can start cutting risk quickly and cheaply, while clinical capacity scales up.
- **The premature-mortality target is within reach.** The combined package bends Indonesia's probability of death from CVD before age 70 down by about 31% by 2050.

Source: results contract (09_deck_results.rds); cascade workbook. Method: interpretation of modelled deaths averted, premature-mortality reduction, and cost per death averted; policy translation needs judgement on feasibility, financing, and delivery. Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

What these results can and cannot tell us

- Results depend on **modelled epidemiological estimates** (GBD) and calibration; the real burden may differ.
- **Effect sizes, future coverage, and unit costs are uncertain**; interventions are assumed to reach and sustain their targets.
- The disease model is **simplified** (healthy → living with CVD → death; no recovery; no comorbidity interactions).
- Value for money is expressed as **cost per death averted**; effects on illness (morbidity) and productivity are not valued in money here.
- **Costing results are preliminary**. Unit costs, service use, and delivery assumptions require further validation against **BPJS and Ministry of Health** data before policy use.
- Projections are **scenarios, not forecasts or guarantees**; they show what is achievable if interventions are delivered as modelled.

*Source: Indonesia CVD model and its input workbooks (GBD 2023; UN WPP 2024; FAIR Choices effects; programme and policy unit costs). **Method:** qualitative statement of the assumptions and uncertainties that bound the quantitative results above.*

The case for action: two levers, affordable, and a reachable target

1. Both, together

Clinical + public health averts **~5.1M deaths** by 2050 — more than either alone.

2. Prevention is affordable

Population policies avert deaths at about **US\$207** each.

3. A reachable target

The probability of death from CVD before age 70 falls **31%** by 2050 under the combined package.

Combined action against CVD could avert about **5.1M deaths** at about **US\$19,299 per death averted** through 2050.

Source: results contract (09_deck_results.rds). Method: deaths averted cumulative 2025–2050; reduction in the probability of death from CVD before age 70 in 2050; cost per death averted uses discounted additional cost. Preliminary costing results: unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

Technical appendix

Appendix 1 — Scenario crosswalk

Family	Workbook scenario id	Executive label
Clinical (FAIR Choices)	all	All selected interventions (combined)
	I_CVD_PRIMARY	Primary prevention of CVD
	I_HF_ACUTE	Acute heart-failure management
	I_HF_CHRONIC	Chronic heart-failure management
	I_IHD_SECONDARY	Secondary prevention of ischemic heart disease
	I_RHD_SECONDARY	RHD secondary prevention
	I_RHD_SURGERY	RHD cardiac surgery
	I_STROKE_SECONDARY	Secondary prevention of stroke
	I_T2D_TREATMENT	Longitudinal treatment of type 2 diabetes
	baseline	Baseline (no new intervention)
Public health	all_public_health	All selected public-health interventions (combined)
	I_PH_SALT_POLICY	Salt policy package
	I_PH_TOBACCO_POLICY	Tobacco policy package
	I_PH_SALT_ENV	Supportive public-institution environment
	I_PH_SALT_FOPL	Sodium front-of-pack labelling
	I_PH_SALT_MEDIA	Sodium mass-media campaign
	I_PH_SALT_REFORM	Mandatory sodium reformulation
	I_PH_TOBACCO_TAX	Tobacco tax
	I_PH_TOB_AD_BAN	Tobacco advertising ban
	I_PH_TOB_CLEAN_AIR	Smoke-free / clean-air law
	I_PH_TOB_MEDIA	Tobacco mass-media campaign
	all_clinical_public_health	All clinical + public-health interventions (combined)
Combined		

Appendix 2 — Definitions, horizons and assumptions

Metrics

- **Deaths averted (cumulative)**
$$= \sum_{t=2025}^{2050} (\text{baseline deaths}_t - \text{scenario deaths}_t),$$

across modelled CVD & diabetes causes.
- **% reduction in deaths, 2050**
$$= 100 \times (\text{baseline} - \text{scenario}) / \text{baseline in 2050}.$$
- **Probability of death from CVD before age 70** =
probability of dying of CVD between exact ages 30 and 70 (six CVD causes, both sexes, single-year life table).
- **% reduction in that probability, 2050:** analogous to deaths.

Model. Discrete-time state-transition model; single-year ages 0–95; six CVD causes (IHD, ischemic stroke, intracerebral haemorrhage, hypertensive heart disease, rheumatic heart disease, cardiomyopathy & myocarditis) plus type-2 diabetes; calibrated to GBD 2023; demography and life expectancy from UN WPP 2024. **Data provenance:** every ordinary-run number is read from the current-run results contract (output/09_deck_results.rds) and the workbooks' literal probability-of-death series; the cascade strategy is read from its isolated workbook; scenario identity follows the Scenario_Catalog contract (binding include_flag). No Excel recalculation is used.

Costing & value for money

- Horizon 2025–2050; cost discount rate 3%/yr; price year 2023.
- Costs in **market US\$**; cost perspective: health system.
- **Cost per death averted** = cumulative **discounted** additional (incremental) cost ÷ cumulative **undiscounted** deaths averted.
- A “Dominant” scenario averts deaths while lowering cost; “not defined” means no deaths averted.

Appendix 3 — How costs are calculated

Intervention-based costing adapted from the DCP Cost Model of Watkins et al. (2020), from a health system perspective with annual projections. For intervention i , sex/age group s , year t :

$$C_{i,s,t} = PIN_{i,s,t} \times cov_{i,s,t} \times f_i \times uc_i$$

$$\Delta C_{s,t} = C_{s,t} - C_{0,t}$$

$$PV(\Delta C_s) = \sum_{t=2026}^{2050} \frac{\Delta C_{s,t}}{(1+r)^{t-2025}}$$

$$\text{Cost per death prevented}_s = \frac{PV(\Delta C_s)}{\sum_{t=2026}^{2050} (D_{0,t} - D_{s,t})}$$

PIN = people in need (population $\times$ epidemiological need); cov = coverage (baseline or scenario); f_i = services per person per year; uc_i = Indonesia-adjusted unit cost; subscript 0 = baseline (current course), s = scenario; r = 3%/yr discount rate; price year 2023; currency market US\$; perspective health system. Deaths D remain **undiscounted** in the cost-per-death calculation. Downstream cost offsets are **excluded**.

Source: adapted from Watkins et al. (2020); unit costs, coverage, and people-in-need from the input workbooks and *AnnualL_Cost*; demography from UN WPP 2024. **Method:** annual costs by intervention $\times$ group $\times$ year, differenced against baseline, discounted to the 2025 base year. **Preliminary costing results:** unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.

Appendix 4 — References

- GBD 2023 Cardiovascular Diseases and Risks Collaborators. 2025. “Global, Regional, and National Burden of Cardiovascular Diseases and Risk Factors in 204 Countries and Territories, 1990–2023.” *Journal of the American College of Cardiology* 86 (22): 2167–2243.
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